Quantum Gravity as Torsional Mismatch in Biquaternion Double Spacetime

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Abstract

General relativity successfully describes gravity as the curvature of a single spacetime continuum, yet its reconciliation with quantum mechanics remains elusive, plagued by singularities, the continuum hypothesis, and the absence of a microscopic origin for curvature. This paper develops the quantum sector of the Double Spacetime Theory (DST) introduced in the companion work, where spacetime is not a shared manifold but a pair of compactified Minkowski spacetimes intrinsically attached to every massive particle and encoded in the 16-real-dimensional biquaternion algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$.

On admissible configurations the torsional scalar obeys $|J| \leq 3\pi^2/8$, or equivalently $|J_{\mathrm{norm}}| \leq 1$. Square-$+1$ generators yield idempotent projectors P_L, P_R and P_{spin}; the naive factors $(1 \pm i)/2$ with $i^2 = -1$ do not, nor does the composite $P_{\mathrm{spin}} P_R$. Usual and dual generators commute on a common axis and need not commute off-axis. The cyclic generators satisfy $IJ = K$ and act on $\mathbb{C}^2$ by $-i\sigma_a$; at every dual rapidity the dual rotor is the Rodrigues matrix of SU(2), which is unitary of determinant one. The grade-1 generators of $\mathrm{Cl}(3,1)$ obey $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ with $\eta = \mathrm{diag}(-1,+1,+1,+1)$, and γ^0 is not the hyperbolic generator j used for P_{spin}. The Killing self-pairing is indefinite and is not a Born probability [1]. The algebras $\mathrm{Cl}(3,1)$ and $\mathrm{Cl}(9,1)$ are not isomorphic, and the dual spinor space is not the Majorana–Weyl module of ten-dimensional spacetime. Variational equivalence of TEGR with the Einstein–Hilbert action is taken from the literature after a rotor-to-tetrad dictionary is fixed. The momentum-space Dirac equation is the first-order factor of the Minkowski quadratic in $\mathrm{Cl}(3,1)$; on the mass shell a dual spinor recovers the rest-frame pair (u, iu), and a longitudinal boost of that pair is the same on-shell construction after a chiral factor along the boost axis. The Euclidean $\mathbb{C}^4$ overlap of two rest solutions that differ by a dual rotation is twice the dual-rotor amplitude, and the same boost multiplies that overlap by $\cosh\alpha$. The mass bilinear $\Psi^\dagger(-i\gamma^0)\Phi$ equals that overlap at rest and is unchanged by the boost. The factor $\cosh\alpha$ is the time component of the Dirac current; the longitudinal component carries $-\sinh\alpha$, the transverse components vanish, and the Clifford contraction of the current with the boosted momentum equals im times the bilinear. The eigenrays of σ_z label the spin along that momentum: the boost preserves each ray, opposite rays carry neither a mass bilinear nor a Dirac current, and a perpendicular dual rotation of angle π exchanges them. These are instances of a covariance under every total rotor: a complex 2×2 matrix M of determinant one, acting by M on the upper Weyl block, by $(M^\dagger)^{-1}$ on the lower and by congruence on $p_0 I + \sum_a p_a \sigma_a$, preserves $Q(p)$, sends solutions of mass m to solutions of mass m, and leaves the mass bilinear and the contraction of the current with a co-transformed vector unchanged. Every total rotor is uniquely a usual rotor followed by a dual rotor. The dual rotors are exactly the total rotors that fix the rest momentum, so the dual sector is the little group of a massive particle, and every momentum on the forward mass shell is reached from rest by exactly one usual rotor. Usual rotors along x and z compose to a usual rotor followed by a nontrivial dual rotation about y, the Thomas–Wigner rotation. For two solutions at one momentum the Dirac current is the momentum with its index raised, times the mass bilinear, so at every on-shell momentum the Euclidean overlap is p_0/m times that bilinear. Under $p_\mu \mapsto -i\partial_\mu$ the momentum-space equation becomes the spacetime Dirac equation $\gamma^\mu \partial_\mu \Psi = -m\Psi$. A plane wave $e^{ip_\mu x^\mu} u$ solves it exactly when u solves the momentum-space equation, and the current of any two solutions of one mass is conserved. Its time component is $i\Psi^\dagger\Psi$, so the density $\Psi^\dagger\Psi$ is nonnegative and conserved; dual rotors leave it unchanged, and a total rotor that carries a rest solution to momentum p multiplies it by p_0/m. This density, not the Killing self-pairing, is what Born's rule requires. A total rotor that acts on the spacetime argument as well as on the spinor carries solutions of the spacetime equation to solutions, and a plane wave of momentum p to the plane wave of the transformed momentum. Minimal coupling $D_\mu = \partial_\mu - ieA_\mu$ to an electromagnetic potential is gauge covariant, a plane wave in a constant potential a solves the coupled equation exactly at the kinetic momentum $p - ea$, and the current stays gauge invariant and conserved. The commutator of covariant derivatives is $-ieF_{\mu\nu}$, so every solution obeys a second-order equation whose field term is $e\,\sigma\cdot(\mathbf{B} + i\mathbf{E})$ on the upper Weyl block and $e\,\sigma\cdot(\mathbf{B} - i\mathbf{E})$ on the lower. The magnetic field acts through the dual generators, identically on both blocks and with gyromagnetic

ratio 2; the electric field acts through the usual generators, with opposite signs on the two blocks. At zero charge the equation is the Klein–Gordon equation. A spin Hamiltonian $\frac{1}{2}\,\omega \cdot \sigma$ evolves a dual spinor by the dual rotor of rapidity $t\omega$, so spin precession in a magnetic field is a dual rotation. When the two electromagnetic phases are equal, interference nodes are ordered by the rapidity, so every common phase that decreases strictly with distance places successive nodes inward; the Bohr radii $n^2 a_0$ increase, and the two sequences do not coincide. The written opposite-sign action frees the mismatch and sources the common rapidity. Affine lag integrates to a cubic common rapidity, and the Jacobi integral along that motion reduces to initial data. The integral is indefinite, so conservation does not restore a frequency. The harmonic stiffness does not select a misalignment. Measurement and the path integral remain geometric hypotheses.

1 Introduction: An Algebraic Approach to Quantum Gravity

The reconciliation of general relativity with quantum mechanics remains one of the most profound open problems in theoretical physics. Despite decades of effort, no consensus framework has emerged that is simultaneously consistent with the empirical successes of both theories while resolving their conceptual incompatibilities. Loop quantum gravity discretizes geometry but retains a background manifold and curvature as fundamental; string theory introduces extra dimensions and supersymmetry without direct observational support; asymptotic safety and causal dynamical triangulations struggle with the continuum limit and the origin of spacetime itself. All existing approaches share a common foundational assumption: the spacetime continuum hypothesis. Spacetime is presumed to exist as a smooth, differentiable manifold independent of matter, with gravity arising from its curvature sourced by energy-momentum.

This assumption, while extraordinarily successful at macroscopic scales, leads to severe difficulties at the quantum level: ultraviolet divergences on curved backgrounds, the black hole information paradox, cosmological singularities, and—most fundamentally—the absence of any physical mechanism explaining why or how matter should curve spacetime. Mach's principle remains unfulfilled; spacetime behaves as an autonomous entity rather than a relational structure determined by the distribution of matter.

In this paper we develop a different approach. The underlying algebra is the 16-real-dimensional biquaternion algebra isomorphic to $\mathrm{Cl}(3,1)$; a background manifold and Riemann curvature are not required for the identities recorded below. The physical picture is the pair of rotors R_{usual} and R_{dual}. Dual-sector kinematics is the representation on $\mathbb{C}^2$; gravitational phenomenology is torsional mismatch measured by $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$. Identification of J with the teleparallel scalar, and Einstein–Hilbert equivalence, are taken from the TEGR literature after a rotor-to-tetrad dictionary is fixed.

At every dual rapidity the dual rotor is the Rodrigues matrix of SU(2): unitary of determinant 1, with relative dual rotors remaining in the group. This is the matrix representation on $\mathbb{C}^2$ recorded in Section 3, not a Riemannian identification of the rotor manifold with the three-sphere. The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators of $\mathrm{Cl}(3,1)$; they obey $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ and are not the hyperbolic bivectors used for P_{spin}. Born's rule is not the Killing self-pairing: that pairing has signature $(3,3)$ and is negative on pure elliptic configurations [1]. Position operators and mass as torsion are not derived here. The momentum-space Dirac equation follows from the Clifford square of the Minkowski quadratic, as recorded in Section 5. It is covariant under every total rotor, and every total rotor is uniquely a usual rotor followed by a dual rotor. The dual factor is the little group of the rest momentum, so spin is carried by the dual sector and momentum by the usual sector. Replacing p_μ by $-i\partial_\mu$ turns the momentum-space equation into the Dirac equation on spacetime. Its on-shell plane waves carry the de Broglie phase, the current of its solutions is conserved, and the time component of that current is the nonnegative density that Born's rule requires. The spacetime equation stays covariant when total rotors act on spacetime points as well, and its minimal coupling to an electromagnetic potential is gauge covariant. In the coupled equation the magnetic field acts on the spinor through the dual generators and the electric field through the usual generators, so the magnetic moment belongs to the dual sector and spin precession in a magnetic field is a dual rotation.

By abandoning the continuum hypothesis entirely, the theory automatically resolves the singularities of classical general relativity. The compactness of the dual spacetime discretizes the rotor angles. On admissible configurations the algebraic ceiling is $|J| \le 3\pi^2/8$ (equivalently $|J_{\mathrm{norm}}| \le 1$); this forbids the unbounded curvature that would otherwise appear at $r = 0$ or at the Big Bang, provided the configuration remains admissible. Event horizons are likewise absent: light rays traverse finite torsional layers via resonance tunneling. Strong-field deviations from general relativity become observable in principle. Dual-only kinematics on the admissible cone delimit when J can be shielded or reversed, and that the dual budget may be routed across silent axes; whether laboratory fields supply that dual force

is the experimental question of Section 11.

The remainder of the paper is organized as follows. Section 2 constructs square-+1 projectors and shows that the composite $P_{\text{spin}}P_R$ is not idempotent. Section 3 identifies dual generators with $-i\sigma_a$ and the dual rotor at every rapidity with an SU(2) matrix, and shows that the Killing pairing is not Born's rule. Section 5 records the Clifford relations of $\mathrm{Cl}(3,1)$, distinguishes γ^0 from the hyperbolic generator j, factors the Minkowski quadratic into the momentum-space Dirac equation, assembles on-shell solutions from a dual spinor, identifies a longitudinal boost of the rest pair with that construction, separates the invariant mass bilinear from the Dirac current, and records that the eigenrays of σ_z label the spin along the momentum; it then establishes covariance under the whole total-rotor group, factors every total rotor uniquely into usual and dual rotors, identifies the dual rotors with the little group of the rest momentum, and obtains the Thomas–Wigner rotation as the dual factor of two perpendicular usual rotors; finally it derives the Gordon identity for the current, passes to the spacetime Dirac equation, and identifies its conserved current with the Born density. The same section extends the total-rotor covariance to spacetime, couples the equation minimally to an electromagnetic potential, and separates the magnetic coupling through the dual generators from the electric coupling through the usual generators. Section 6 records that equal-scale interference nodes run inward for every common phase that decreases strictly with distance, and that this sequence is distinct from the outward Bohr radii. Sections 7–8 record that the harmonic stiffness does not select a misalignment, identify spin precession with dual-rotor evolution, and discuss measurement and the path integral as geometric hypotheses; the written particle action is shown to free the mismatch and source the common rapidity. Section 9 presents conjectural predictions bounded by the admissible ceiling. Section 10 compares the framework with other programmes, distinguishing internal duality from ten-dimensional spacetime supersymmetry. Section 11 outlines near-term tests. Section 12 gathers the algebraic picture of the quantum sector. Section 13 summarizes.

Quantum gravity is not claimed to follow in full from $\mathrm{Cl}(3,1)$. The results of this paper are algebraic identities in the 16-dimensional algebra; continuum dynamics remains a classical approximation.

2 Biquaternion Spinors and the Minimal Left Ideal

The biquaternion algebra $\mathbb{H} \oplus \mathbb{H}$, canonically isomorphic to the Clifford algebra $\mathrm{Cl}(3,1)$ of Minkowski spacetime with signature $(-,+,+,+)$, provides a natural and minimal realization of Dirac spinors. Unlike conventional approaches that introduce spinors as an external representation space, here the spinor emerges directly as an element of the algebra itself.

A primitive idempotent that projects onto an irreducible representation must be built from generators that square to $+1$. If g satisfies $g^2 = +1$, then

$$\frac{1+g}{2}$$

is idempotent. Write $j := e_0 e_1$ for the hyperbolic generator of axis 0 (so $j^2 = +1$). This j is *not* the time vector $\gamma^0 = e_0$, which squares to -1 in signature $(-,+,+,+)$ (Section 5). The working projectors are

$$P_{\text{spin}} = \frac{1+j}{2} = \frac{1+e_0e_1}{2}$$

and the complementary spatial chirality pair

$$P_L = \frac{1-e_1}{2}, \qquad P_R = \frac{1+e_1}{2}$$

(with $e_1^2 = +1$). They satisfy $P_L^2 = P_L$, $P_R^2 = P_R$, and $P_L + P_R = 1$.

The product

$$P = P_{\text{spin}}\, P_R$$

is not idempotent: e_0e_1 and e_1 anticommute, the factors do not commute, and $(P_{\text{spin}}P_R)^2 \neq P_{\text{spin}}P_R$. A commuting square-+1 pair is e_1 with the off-axis boost e_0e_2; their product projector is idempotent and generates a left ideal of real dimension eight. That ideal is the direct sum of the four-dimensional Minkowski core $\mathrm{Cl}(3,1)\,P$ and a copy along the null generator e_4; the copy is a proper $G(3,1,1)$-submodule, so the eight-dimensional ideal is reducible as a PGA-module. The match with the real dimension of a Dirac spinor $\mathbb{C}^4$ is this null doubling, not an identification with an irreducible Dirac module. Irreducibility of the Minkowski core as a $\mathrm{Cl}(3,1)$-module is not claimed.

By contrast, the naive formulae built from the $\mathrm{Cl}(3,1)$ pseudoscalar i alone,

$$\frac{1 \pm i}{2} \qquad (i^2 = -1),$$

are not idempotent: $\big((1-i)/2\big)^2 = -i/2$. Composites such as $i\gamma_1\gamma_2$ that square to $+1$ remain admissible building blocks; the bare pseudoscalar does not.

A general element of a candidate spinor space takes the explicit form

$$\psi = \big(a_0 + a_1 i + b_1 kI + b_2 kJ + b_3 kK + c_1 jI + c_2 jJ + c_3 jK\big)P,$$

where $a_0, a_1, b_i, c_i \in \mathbb{R}$ and P is a square-$+1$ projector (not $P_{\text{spin}}P_R$). Identifying the pseudoscalar i with the imaginary unit of quantum mechanics is a dictionary, not a proof that the carrier is the Dirac module. The dual map $X \mapsto Xi$ still reverses time orientation; the algebraic projectors themselves are P_L and P_R above, not $(1 \pm i)/2$.

The action of rotors on algebra elements is left multiplication. Same-axis usual and dual generators commute; off-axis pairs need not. In particular Γ_a does not commute with every usual-sector generator $i\Gamma_b$, so the order of the factors in $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$ matters off-axis. With that order fixed the factorization is an identity in the $\mathbb{C}^2$ representation: every total rotor is a usual rotor followed by a dual rotor, and both factors are unique (Section 5).

The bilinear $\bar{\psi}\psi \propto \mathrm{Tr}(\Omega_{\text{biv}})$ and the emergence of the full fermionic Standard Model from the 16-dimensional algebra are not established here. What follows from the algebra is the projector calculus above.

3 Quantum States as Dual-Spacetime Rotations

The dual rotor

$$R_{\text{dual}} = \exp\left(\sum_{a=1}^{3} \frac{\phi_a}{2}\Gamma_a\right),$$

where $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$ satisfy the quaternion relations $\Gamma_a^2 = -1$ and $IJ = K$, is represented on $\mathbb{C}^2$ by $-i\sigma_a$. The alternative dictionary $I \leftrightarrow i\sigma$ is incompatible with $IJ = K$.

At every dual rapidity β the exponential is the Rodrigues formula

$$U(\beta) = \cos(\|\beta\|/2)\, I + \sin(\|\beta\|/2)\, n \cdot (-i\sigma),$$

with $n = \beta/\|\beta\|$ when $\beta \neq 0$ and $U(0) = I$. That matrix is unitary of determinant 1. Relative dual rotors remain in $\mathrm{SU}(2)$. The inverse is the opposite rapidity, so a relative dual rotor is a product in the group: its character records the alignment of the two dual axes, and the noncommuting residue is a traceless tilt. A 2π dual rotation about a computational axis is $-I$. Compactness of the dual spacetime ($\phi_a \bmod 4\pi$) is compatible with this matrix group on $\mathbb{C}^2$; an independent diffeomorphism of manifolds $S^3 \cong \mathrm{SU}(2)$ is not proved here.

Position operators, the canonical commutation relations, and the uncertainty bound $\Delta\phi \cdot \Delta\theta \geq 1/2$ are geometric hypotheses. Momentum enters spacetime only through the correspondence $p_\mu \mapsto -i\partial_\mu$ of Section 5.

The probabilistic interpretation is not the Killing self-pairing. The identification

$$|\psi|^2 \propto \tfrac{1}{2}B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$$

fails: the Killing pairing has signature $(3,3)$ and is negative on pure elliptic configurations. Self-overlap is therefore not a probability. Born's rule belongs to the Hilbert layer and is not a consequence of J [1]. The nonnegative conserved density it requires is the time component of the Dirac current, $\Psi^\dagger\Psi$ (Section 5).

Measurement, entanglement, and collapse as torsional synchronization are likewise left open. Dual-sector kinematics in this paper means the $\mathbb{C}^2$ representation of the cyclic generators.

4 Dual Spacetime as Intrinsic Supersymmetry

The cosmological constant problem, widely known as the vacuum catastrophe, represents one of the most acute fine-tuning crises in theoretical physics. Quantum field theory predicts that the vacuum

energy density arises from the zero-point fluctuations of every field mode up to the ultraviolet cutoff. With the cutoff taken at the Planck scale, this yields an energy density of order $M_{\mathrm{Pl}}^4 \approx 10^{74}\,\mathrm{GeV}^4$, which converts in SI units to approximately $10^{113}\,\mathrm{J/m}^3$. The observed value, however, inferred from the present-day accelerated expansion, is only $\rho_{\mathrm{vac}}^{\mathrm{obs}} \approx 5.4 \times 10^{-10}\,\mathrm{J/m}^3$, corresponding to a cosmological constant $\Lambda \sim 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$. The ratio between the theoretical prediction and the measured value therefore reaches

$$\frac{\rho_{\mathrm{vac}}^{\mathrm{theory}}}{\rho_{\mathrm{vac}}^{\mathrm{obs}}} \sim 10^{120}.$$

No known renormalization procedure can remove this discrepancy without an extraordinary cancellation between bare parameters and quantum corrections, accurate to one part in 10^{120}. The problem is not technical but conceptual: gravity cannot be consistently coupled to the vacuum of quantum field theory on a fixed background without destroying the observed cosmology.

The Double Spacetime Theory resolves the crisis by revealing that supersymmetry is already realized *intrinsically and inseparably* inside the 16-real-dimensional biquaternion algebra carried by every massive particle. No new fields or particles are introduced, and no external symmetry is imposed or broken.

The six generators of the underlying Lorentz candidate $\mathfrak{so}(3,1)$ (not the twelve-dimensional sum $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$) partition into two sectors distinguished by the sign of their quadratic form: the hyperbolic sector, whose generators square to $+1$, and the cyclic sector, whose generators square to -1. Same-axis generators commute; off-axis commutators need not vanish. The complete rotor R_{total} acts on both sectors simultaneously. The dual map $X \mapsto Xi$ interchanges the two sectors up to signs while reversing time orientation. This interchange constitutes an internal automorphism of the algebra itself; it does not map a particle onto a distinct species but merely exchanges two permanently coupled aspects of one and the same 16-dimensional object. Consequently, the supersymmetric partner of any bosonic excitation is already present as the dual-sector component. There is no external superpartner to discover.

The scalar J is the parameter-space quadratic $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ built from the mismatch bivector Ω_{biv}. Because the Killing form takes opposite values on the hyperbolic sector $(+8)$ and the cyclic sector (-8), bosonic and fermionic contributions enter with opposite signs. When the two sectors are aligned, these contributions cancel exactly. The residual vacuum value of J is generated exclusively by quantum fluctuations of the dual-sector angles. The cyclic generators square to -1, rendering the dual spacetime elliptic: its rotor manifold is the compact three-sphere $S^3 \cong \mathrm{SU}(2)$. The angles are therefore periodic and discretely sampled, $\theta_a \in (2\pi/N)\mathbb{Z}$ for large integer N. Compactness does *not* force an algebraic upper bound of order N^{-2}. On admissible configurations $|J| \le 3\pi^2/8$, or $|J_{\mathrm{norm}}| \le 1$ with $J_{\mathrm{norm}} = \frac{8}{3\pi^2}J$. The discrete quantity $\varepsilon_N = 16/(3N^2)$ is a *lower* bound on the nonzero normalized height, not an upper bound: every non-rest admissible lattice point satisfies $|J_{\mathrm{norm}}| \ge \varepsilon_N$. Vacuum suppression, if it occurs, must come from a dynamical cancellation between sectors, not from writing $|J| \le J_{\max} \sim N^{-2}$.

Because the dual rotor is the SU(2) Rodrigues matrix at every rapidity, the dual sector can host spin-1/2 kinematics; a full identification of Standard Model fermions with dual phases is not claimed. The dual map $X \mapsto Xi$ is an internal automorphism of the algebra. It is not spacetime supersymmetry of a ten-dimensional string: $\mathrm{Cl}(3,1) \ncong \mathrm{Cl}(9,1)$, the dual spinor space is not the Majorana–Weyl module of $\mathrm{Cl}(9,1)$, and the Super-Poincaré generator count is not the ten generators of the projective geometric algebra.

The opposing signatures cancel when the sectors align. That cancellation does not generate the observed fermion masses, and the admissible bound on J is not the electroweak scale.

In this framework supersymmetry is not an optional extension that may or may not be realized in nature. It is the inevitable geometric consequence of an algebra whose two sectors possess quadratic forms of opposite signature and whose dual sector is compact. The vacuum catastrophe is not solved by adding new symmetries; it is dissolved by recognizing that the 16-dimensional biquaternion structure already contains, in its opposing signatures and intrinsic compactness, the exact cancellation required by observation.

5 Clifford relations and the Dirac matrices

The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators γ^μ of $\mathrm{Cl}(3,1)$. They obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1,+1,+1,+1).$$

In particular $(\gamma^0)^2 = -1$. The hyperbolic generator $j = e_0 e_1$ used for P_{spin} squares to $+1$, so $\gamma^0 \neq j$. The dictionary $\gamma^0 \leftrightarrow j$, $\gamma^i \leftrightarrow kI, \ldots$ is therefore replaced by this grade-1 basis.

For a four-momentum p the grade-1 element $\gamma(p) = \sum_\mu p_\mu \gamma^\mu$ squares to the Minkowski quadratic $Q(p) = -p_0^2 + p_1^2 + p_2^2 + p_3^2$, both in $\mathrm{Cl}(3,1)$ and as matrices on $\mathbb{C}^4$. The two chiral slashes of the Weyl representation multiply to that same quadratic. The mass shell $Q(p) = -m^2$ is therefore the vanishing of $(\gamma(p) - im)(\gamma(p) + im)$. A spinor with $\gamma(p)\Psi = im\Psi$ lies on that shell: this is the Dirac equation in momentum space. In the Weyl representation $\gamma(p)$ is off-diagonal, so vanishing mass decouples the two chiral blocks while a nonzero mass mixes them. Chirality γ^5 sends a solution of mass m to a solution of mass $-m$.

On the mass shell with $m \neq 0$ an arbitrary dual spinor u determines a solution: the upper Weyl block is u, and the lower block is fixed by the chiral slash of p. At rest momentum $(m, 0, 0, 0)$ this is the pair $\Psi = (u, iu)$. Dual rotation of u remains a solution at the same four-momentum: the dual rotor acts internally on the Weyl parameter.

A longitudinal boost of rest momentum,

$$p = (m\cosh\alpha,\ 0,\ 0,\ m\sinh\alpha),$$

stays on the shell. At that momentum the Weyl construction is the Dirac-boosted rest-frame pair. The boost acts by the hyperbolic Rodrigues factors

$$V(\alpha) = \cosh(\alpha/2)\, I + \sinh(\alpha/2)\, \sigma_z$$

on the upper Weyl block and $V(-\alpha)$ on the lower: the rest pair (u, iu) is sent to the on-shell solution whose Weyl parameter is $V(\alpha)u$. Dual rotation about the boost axis commutes with $V(\alpha)$, so internal dual rotation and the longitudinal boost may be taken in either order. Dual rotation about a perpendicular axis does not commute with $V(\alpha)$.

The $\mathbb{C}^4$ overlap of two rest-frame solutions that differ by a dual rotation is twice the dual-rotor amplitude. A longitudinal boost of rapidity α multiplies that overlap by $\cosh\alpha$, for every dual rotation. The boosted overlap therefore agrees with the rest-frame overlap only at vanishing rapidity, so the overlap is not invariant under the boost. It remains a Euclidean identity on $\mathbb{C}^4$, not a spacetime inner product.

The mass bilinear of two rest-frame solutions, assembled from arbitrary dual spinors u and v, is $\bar\Psi\Phi = \Psi^\dagger(-i\gamma^0)\Phi$. At rest it coincides with the Euclidean overlap, and therefore with twice the dual-spinor pairing. A longitudinal boost leaves $\bar\Psi\Phi$ unchanged, so the two pairings agree for every such pair only at vanishing rapidity. The Dirac current built from the same matrix has components

$$j^\mu = \Psi^\dagger(-i\gamma^0)\gamma^\mu\Phi = i\,\bar\Psi\Phi\,(\cosh\alpha,\ 0,\ 0,\ -\sinh\alpha).$$

In this representation $(-i\gamma^0)\gamma^0 = i$, and the time component is i times the Euclidean overlap: the factor $\cosh\alpha$ belongs to that component. The transverse components vanish. The Clifford contraction of the current with the boosted momentum equals $im\bar\Psi\Phi$, and $mj^0 = i\bar\Psi\Phi\, p^0$, $-mj^3 = i\bar\Psi\Phi\, p^3$. For $m \neq 0$ the combination $(j^0, -j^3)$ is parallel to p.

The eigenrays of σ_z label the spin of this solution along the momentum. $V(\alpha)$ multiplies the eigenvalue $+1$ by $e^{\alpha/2}$ on the upper block and by $e^{-\alpha/2}$ on the lower, and both blocks remain in that ray. The eigenvalue -1 exchanges the two factors, so at large positive rapidity the first lies mostly in the upper block and the second mostly in the lower. The invariant mass bilinear between the two rays vanishes at every longitudinal rapidity, and so does every component of the Dirac current between them. A dual rotation about the boost axis multiplies the eigenvalue $+1$ by $e^{-i\theta/2}$ and the eigenvalue -1 by $e^{i\theta/2}$. A dual rotation of angle π about a perpendicular axis exchanges the two rays.

The longitudinal boost is one instance of a covariance under the whole total-rotor group. In the $\mathbb{C}^2$ representation a total rotor is a complex matrix M of determinant one. A four-momentum is the Hermitian matrix $\hat p = p_0 I + \sum_a p_a \sigma_a$, whose determinant is $-Q(p)$, and M acts on it by $\hat p \mapsto M\hat p M^\dagger$; on a Dirac spinor it acts by M on the upper Weyl block and by $(M^\dagger)^{-1}$ on the lower. This action preserves $Q(p)$ and intertwines both chiral slashes, so it sends every solution of mass m at p to a solution of the same mass at the transformed momentum. The mass bilinear $\bar\Psi\Phi$ is invariant, and the Clifford contraction of the Dirac current with a four-vector is unchanged when the vector is transformed together with the spinors. A dual rotor is unitary and acts by the same matrix on both blocks. A usual rotor $V(\alpha) = \exp(\alpha \cdot \sigma/2)$ is Hermitian and acts on the lower block by its inverse, as $V(-\alpha)$ does for the longitudinal boost.

Every total rotor is a usual rotor followed by a dual rotor, $M = V(\alpha)U(\beta)$, and both factors are unique: $V(\alpha)$ is the positive square root of $MM^\dagger$, and $U(\beta) = V(\alpha)^{-1}M$. A total rotor fixes the rest

momentum $(m,0,0,0)$ with $m \neq 0$ exactly when it is a dual rotor. The dual sector is therefore the little group of a massive particle, and its spin degrees of freedom are those of the dual spinor u. The momentum is carried by the usual factor. The dual factor leaves the rest momentum in place, and every momentum on the forward mass shell is reached from rest by exactly one usual rotor. Usual rotors along perpendicular axes do not commute. Their product does not reduce to a usual rotor: the product of usual rotors along x and z is a usual rotor followed by a nontrivial dual rotation about y, the axis perpendicular to both. This dual factor is the Thomas–Wigner rotation. In double spacetime it is an internal rotation of the dual spinor, produced by composing usual rotors.

The current of two solutions at one momentum is fixed by that momentum. The adjoint matrix $-i\gamma^0$ is Hermitian and every $(-i\gamma^0)\gamma^\mu$ is anti-Hermitian, so the anticommutator $\{\gamma(p),\gamma^\mu\} = 2p^\mu$ yields the Gordon identity

$$m\, j^\mu = -i\, p^\mu\, \bar{\Psi}\Phi, \qquad p^\mu = \eta^{\mu\nu} p_\nu,$$

for any two solutions Ψ and Φ of mass m at p. The longitudinal current above is its instance along z. The time component reads $m\,\Psi^\dagger\Phi = p_0\,\bar{\Psi}\Phi$: at every on-shell momentum the Euclidean overlap is p_0/m times the invariant bilinear, and $\cosh\alpha$ is that ratio for the longitudinal boost. For a single spinor the current is i times a real four-vector J^μ with $J^0 = \Psi^\dagger\Psi$. On a nonzero solution with $m \neq 0$ the component p_0 does not vanish, $\bar{\Psi}\Psi = (m/p_0)\,\Psi^\dagger\Psi$ is real and nonzero, and J has Minkowski square $-(\bar{\Psi}\Psi)^2$. The current is therefore timelike and parallel to the momentum, and its invariant length is the mass bilinear. With $m > 0$ and $p_0 > 0$ that bilinear is positive and does not exceed $\Psi^\dagger\Psi$. A total rotor that carries the rest momentum to p multiplies the density $\Psi^\dagger\Psi$ of a rest solution by p_0/m, while every dual rotor leaves it unchanged. The density is fixed by the little group and changes only with the usual factor.

Replacing p_μ by $-i\partial_\mu$ turns the momentum-space equation $\gamma(p)\Psi = im\Psi$ into the Dirac equation on spacetime,

$$\gamma^\mu \partial_\mu \Psi = -m\Psi.$$

A plane wave $\Psi(x) = e^{ip_\mu x^\mu} u$ solves it exactly when its amplitude u solves the momentum-space equation at p. The solutions of one mass form a complex vector space, so every finite superposition of on-shell plane waves of mass m is again a solution. The phase $p_\mu x^\mu$ is the de Broglie phase, and the amplitude is the on-shell spinor assembled from a dual spinor. For any two differentiable solutions of the same real mass the current $j^\mu = \bar{\Psi}\gamma^\mu\Phi$ is conserved,

$$\partial_\mu j^\mu = 0,$$

and between two plane waves of one momentum it is uniform. For $\Phi = \Psi$ the time component $j^0 = i\Psi^\dagger\Psi$ makes $\rho = \Psi^\dagger\Psi$ a density that is nonnegative, vanishes only where Ψ does, and obeys the continuity equation $\partial_t\rho + \nabla\cdot J = 0$ with the real spatial current $J^k = -ij^k$. This is the density that Born's rule requires. It is carried by the Dirac current, not by the Killing self-pairing, which is indefinite.

The total-rotor covariance of the momentum-space equation extends to spacetime. Let $\Lambda(M)$ be the action of a total rotor on four-momenta described above, and let $S(M)$ be its action on Dirac spinors, M on the upper Weyl block and $(M^\dagger)^{-1}$ on the lower. The field

$$\Psi'(x) = S(M)\,\Psi\big(\Lambda(M)^{\mathsf{T}} x\big)$$

solves $\gamma^\mu\partial_\mu\Psi' = -m\Psi'$ whenever Ψ does. The transposed action on the argument is fixed by the phase: $p_\mu x^\mu$ evaluated at $\Lambda(M)^{\mathsf{T}}x$ is $(\Lambda(M)p)_\mu x^\mu$, so the plane wave $e^{ip_\mu x^\mu}u$ goes to the plane wave of momentum $\Lambda(M)p$ with amplitude $S(M)u$. The de Broglie phase is transported by the usual factor; a dual rotor fixes the rest momentum and leaves the phase of a rest-frame plane wave in place.

An electromagnetic potential A_μ enters through the covariant derivative $D_\mu = \partial_\mu - ieA_\mu$, and the coupled equation is

$$\gamma^\mu D_\mu \Psi = -m\Psi.$$

It is gauge covariant: for every differentiable χ, the pair $\Psi \mapsto e^{ie\chi}\Psi$, $A_\mu \mapsto A_\mu + \partial_\mu\chi$ carries solutions to solutions. The gauge factor is a scalar U(1) phase. It commutes with every total rotor and is neither a dual rotor nor its character. In a constant potential a the plane wave $e^{ip_\mu x^\mu}u$ solves the coupled equation exactly when u solves the momentum-space equation at the kinetic momentum $p - ea$. The canonical momentum p carries the phase; the kinetic momentum carries the mass shell and the spinor assembled from a dual spinor. The current $\bar{\Psi}\gamma^\mu\Phi$ is unchanged by a gauge transformation of both spinors, and for real mass, charge and potential it remains conserved on solutions of the coupled equation. The Born density therefore survives the coupling.

For twice-differentiable spinors the covariant derivatives fail to commute by the field strength,

$$[D_\mu, D_\nu]\Psi = -ieF_{\mu\nu}\Psi, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Applying $\gamma^\nu D_\nu$ to the coupled equation and splitting $\gamma^\mu\gamma^\nu$ into its symmetric and antisymmetric parts, every twice-differentiable solution obeys

$$\eta^{\mu\nu} D_\mu D_\nu \Psi - \frac{ie}{2} F_{\mu\nu}\gamma^\mu\gamma^\nu \Psi = m^2 \Psi.$$

At zero charge this is the Klein–Gordon equation $\eta^{\mu\nu}\partial_\mu\partial_\nu\Psi = m^2\Psi$ for every component of a free solution. The field-strength term is block diagonal in the Weyl representation. With the magnetic field $\mathbf{B} = (F_{23}, F_{31}, F_{12})$ and the electric field $E_k = F_{k0}$, it acts as

$$e\,\sigma\cdot(\mathbf{B} + i\mathbf{E}) \text{ on the upper block}, \qquad e\,\sigma\cdot(\mathbf{B} - i\mathbf{E}) \text{ on the lower block}.$$

The magnetic part is the same on both blocks. Since $\sigma\cdot\mathbf{B} = i\sum_a B_a(-i\sigma_a)$ and $-i\sigma_a$ are the dual generators, the magnetic field acts on the spinor through the dual sector, in the way a dual rotor acts by one matrix on both blocks. The electric part changes sign between the blocks, as the generators of a usual rotor do, which acts on the lower block by its inverse. The magnetic field therefore couples to the spin carried by the little group, and the electric field to the boost carried by the usual sector. With $\Pi_k = -iD_k$ the spatial derivatives and the magnetic term combine into $-(\sigma\cdot\Pi)^2$, because $(\sigma\cdot\Pi)^2 = \Pi^2 - e\,\sigma\cdot\mathbf{B}$ follows from the commutator. The magnetic moment thus enters with gyromagnetic ratio 2, the value fixed by the first-order equation.

Square-+1 projectors $P_L = (1 - e_1)/2$ and $P_R = (1 + e_1)/2$ organise a chiral split in the algebra; they do not by themselves yield the Standard Model weak interaction.

6 The bound electron

The electron of Section 5 is free: a dual spinor on the mass shell, the pair (u, iu) at rest. Binding it to a charge sets the two electromagnetic phases equal. The scalar interference factor of that common phase has exactly one zero in each interval of length π in the half-rapidity. Those zeros are force nodes, ordered by the rapidity. Distance enters only by inverting the common phase, so a phase that decreases strictly with distance places the next node inside the previous one, at a radius ratio less than 1.

The Bohr radii $r_n = n^2 a_0$ increase with n, and the first two stand in the ratio 4. No such phase places two successive nodes on two successive members of that sequence. In particular a_0 and $4a_0$ are not the radii of the first two nodes, and the hydrogen spectrum is a different sequence from the equal-scale nodes.

A circular speed $v^2 = ke^2/(m\gamma_s r)$ is real only where the interference factor is positive. A node is a divergence of that force, and the barrier does not quantize the orbit. The equal-scale geometry fixes the inward order of the barriers and the sign of each layer. The outward Rydberg law and the absolute scale of the barriers remain separate inputs.

7 Torsional Stiffness and the Higgs Reading

Section 5 gives the momentum-space Dirac equation as a Clifford factor of $Q(p)$. What follows is a geometric hypothesis for the *value* of m, not a derivation of that value from torsion.

The torsional mismatch between the usual-sector rotor R_{usual} and the dual-sector rotor R_{dual} is quantified by the relative angle $\delta_a = \phi_a - \theta_a$. A modelling stiffness proportional to the rest mass m may be assigned to that angle. In the low-energy limit the associated potential is

$$V_{\text{torsion}} = \frac{1}{2} m \sum_{a=1}^{3} \delta_a^2.$$

This V_{torsion} is a particle-intrinsic oscillator potential, not the parameter-space Killing quadratic of the main paper, and it is not a restoring force in the Euler–Lagrange system of the written opposite-sign action of Section 8. The bilinear

$$m\bar\psi\psi \propto \mathrm{Tr}(\Omega_{\text{biv}}),$$

with $\Omega_{\text{biv}} = \log(R^\dagger_{\text{usual}} R_{\text{dual}})$, is a geometric reading of the Dirac mass, not a derivation of m from the lag.

The harmonic stiffness is not a Higgs mechanism. In the standard model the Higgs field acquires a vacuum expectation value away from the symmetric point, and the curvature there generates masses. Here the written potential is minimized at alignment. Under the opposite kinetic signs of the particle

action every constant lag is free, and the lag is inertial only where the slope vanishes, so a nonzero harmonic lag accelerates the common rapidity rather than settling into a misalignment. A quartic well off the origin can freeze a nonzero lag, but only when the two kinetic signs agree; the common phase is then flat, and its rate is initial data, not a mass for W or Z. No Weinberg angle is produced. On one axis that alignment is $J = \frac{1}{2}\delta\sigma = 0$ with $M = \sigma^2/4$, the vacuum only at the origin, and a frozen well at rest is a balanced state of mass $v^2/4$.

8 Unified Dynamics of Quantum Mechanics and Gravity

The rotor Lagrangian is a model for the six rapidities, not a deduction from unitarity of the rotors. Orbital Schrödinger evolution, measurement as synchronization, entanglement, and the path integral below remain geometric hypotheses; the spin part of Schrödinger evolution is a dual rotor. In that picture, dual-rotor kinematics is the quantum side and usual–dual misalignment is the gravitational side.

Consider a single massive particle whose complete rotor is $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$. The written effective action for the intrinsic rotor angles is

$$S_{\text{particle}} = \int \left[\frac{1}{2} \sum_{a=1}^{3} \dot{\phi}_a^2 - \frac{1}{2} \sum_{a=1}^{3} \dot{\theta}_a^2 - \frac{m}{2} \sum_{a=1}^{3} (\phi_a - \theta_a)^2 \right] dt.$$

The opposite signs of the kinetic terms are part of the written model. They are intended to reflect the boost-like (hyperbolic) character of the usual sector and the rotation-like (elliptic) character of the dual sector; they are not deduced from unitarity of the rotors. The last term is the oscillator potential of Section 7, not the parameter-space Killing quadratic of the main paper.

On one axis the density is equivalently the mixed form

$$L = \tfrac{1}{2}\dot{\sigma}\dot{\delta} - \tfrac{m}{2}\delta^2,$$

with common rapidity $\sigma = \phi + \theta$ and mismatch $\delta = \phi - \theta$. The Euler–Lagrange equations of this density are

$$\ddot{\phi}_a = -m(\phi_a - \theta_a), \qquad \ddot{\theta}_a = -m(\phi_a - \theta_a).$$

Both angles accelerate together. The mismatch is therefore free: $\ddot{\delta}_a = 0$. What the potential sources is the common rapidity, $\ddot{\sigma}_a = -2m\delta_a$. The free mismatch integrates at once. On one axis the general solution is the affine lag $\delta(t) = \delta_0 + \nu t$, and the common rapidity is the cubic

$$\sigma(t) = \sigma_0 + \dot{\sigma}_0\, t - m\delta_0\, t^2 - \frac{m\nu}{3}\, t^3.$$

A frozen lag accelerates σ uniformly; a nonzero rate makes the acceleration itself linear in time. Along this motion the Jacobi integral collapses to the initial value $\frac{1}{2}\nu\dot{\sigma}_0 + \frac{m}{2}\delta_0^2$: the quadratic growth of $\dot{\sigma}$ cancels the quadratic growth of the potential, so the charge stays finite while the rapidity runs away. The mass, written as a stiffness of δ, is a force on σ and does not restore a frequency [4].

The Jacobi integral

$$E = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}\dot{\theta}^2 + \tfrac{m}{2}\delta^2 = \tfrac{1}{2}\dot{\sigma}\dot{\delta} + \tfrac{m}{2}\delta^2$$

is conserved along those equations and is indefinite: a purely dual velocity at vanishing lag already yields $E < 0$. Conservation of an unbounded quadratic does not select a frequency.

A neighbouring system sometimes written alongside the action,

$$\ddot{\phi}_a = m(\phi_a - \theta_a), \qquad -\ddot{\theta}_a = m(\phi_a - \theta_a),$$

makes the mismatch runaway, $\ddot{\delta}_a = 2m\delta_a$. It is not an oscillator.

An oscillator appears only if both kinetic signs are taken positive. The density $\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\dot{\theta}^2 - \frac{m}{2}(\phi - \theta)^2$ yields $\ddot{\delta} + 2m\delta = 0$, at frequency $\sqrt{2m}$ in units $\hbar = c = 1$, while the common rapidity is free. The general solution is $\delta(t) = A\cos(\sqrt{2m}\,t) + B\sin(\sqrt{2m}\,t)$, with $\delta(t)^2 \le A^2 + B^2$, and its Jacobi integral equals $\frac{1}{4}\dot{\sigma}_0^2 + \frac{m}{2}(A^2 + B^2)$. That frequency and the Compton frequency m meet only at the isolated values $m = 0$ and $m = 2$, and the model is not the written action. For $m \ge 0$ its Jacobi integral is nonnegative, so conservation does bound the motion of δ. The neighbouring sign pattern $\ddot{\delta} = 2m\delta$, at the same rate, is instead the unbounded hyperbolic motion $A\cosh(\sqrt{2m}\,t) + B\sinh(\sqrt{2m}\,t)$.

The written action therefore supplies no harmonic bridge to quantum theory. Identifying the rate ν of free lag with $E/\hbar$ remains a reading; the accumulated lag may then be written

$$\int \delta_a \, dt \propto \frac{Et - \mathbf{p} \cdot \mathbf{x}}{\hbar},$$

the de Broglie phase of the matter wave [3]. The same phase carries the plane-wave solutions of the spacetime Dirac equation of Section 5; what remains a reading is its identification with the free lag. The scalar U(1) factor built from δ is distinct from the real character of the dual rotor and from its spinor matrix element.

The Schrödinger equation is readable as the unitary evolution of the dual rotor under an external potential that modulates the dual phases. For the spin this reading is exact. A Hamiltonian $\frac{1}{2}\,\omega\cdot\sigma$, linear in the Pauli matrices, evolves a dual spinor by the dual rotor $U(t\omega)$: the curve $u(t) = U(t\omega)u_0$ solves

$$i\,\dot{u} = \tfrac{1}{2}\,(\omega \cdot \sigma)\,u,$$

and its rapidity grows linearly in time. In the nonrelativistic reduction of the coupled Dirac equation of Section 5, the spin part of the Pauli Hamiltonian in a uniform magnetic field is $-(e/2m)\,\sigma \cdot \mathbf{B}$, of this form with $\omega = -(e/m)\mathbf{B}$. The spin precesses as a dual rotation at the angular frequency $|e|B/m$, which equals the cyclotron frequency of the orbit, as gyromagnetic ratio 2 requires. The orbital part of the Schrödinger equation is not linear in the dual generators. Compactness of the dual sector would keep that evolution unitary. It is not derived from the written particle action.

Wave-function collapse is readable as forced synchronization of dual rotors across an entangled many-particle system. A measurement interaction that coupled to the usual spacetime would still have to supply a restoring mechanism: the written action does not drive δ_a toward zero, and minimizing a global J is a further identification, not an Euler–Lagrange consequence. The apparent non-unitary jump would then be the classical limit of a rapid, collective torsional relaxation. No additional collapse postulate is required in that picture; the picture itself remains a hypothesis.

Quantum entanglement likewise acquires a torsional reading. When two particles share correlated dual phases, their collective mismatch energy is lowered only after a potential for that mismatch is chosen. A local operation on one particle that alters its dual rotor would then change a global J, producing the instantaneous correlation characteristic of entanglement without violating causality (information propagates at most at the speed of light through the usual sector). Compactness of the dual sector makes that non-locality available as a geometric option, not as a theorem of the written action.

A path-integral formulation may be written with the same particle action,

$$\langle R_f | R_i \rangle = \int \mathcal{D}[\phi, \theta] \, \exp\left(\frac{i}{\hbar} S_{\text{particle}}[\phi, \theta]\right),$$

the integral running over paths in the dual-rotor phase space. Compactness of the dual sector would keep the integral well-defined. Because the written S_{particle} makes δ free, the torsional potential term does not by itself generate Compton-frequency interference.

In this picture quantum mechanics and gravity are not separate theories that must be reconciled. Quantum mechanics is the kinematics of rotations in the compact dual spacetime; gravity is the dynamical cost of breaking the parallelism between the usual and dual rotors. Both readings originate from the same 16-dimensional biquaternion algebra. The kinematics is sharp. The dynamical bridge from the written particle action is not.

9 Concrete Predictions of Quantum Gravity

The double spacetime theory yields a series of sharp, falsifiable predictions that distinguish it decisively from both general relativity and other quantum gravity approaches. These predictions follow directly from the compactness of the dual spacetime and the resulting discretization of rotor angles.

9.1 Discrete Spacetime and Singularity Resolution

The dual rotor angles are discretized by the compact topology of the dual spacetime:

$$\phi_a, \theta_a \in \frac{2\pi}{N}\mathbb{Z},$$

where the integer N is determined by the particle's rest mass in Planck units. Consequently, on admissible configurations the torsional mismatch scalar is bounded by

$$|J| \leq \frac{3\pi^2}{8}, \qquad |J_{\text{norm}}| \leq 1.$$

A separate Planck-scale estimate of the form $c^4/G\,\ell_P^{-2}$ is a dimensionful curvature/energy scale and must not be written as an upper bound on the dimensionless algebraic J. The two statements are not interchangeable, and neither is the discrete lower bound $\varepsilon_N = 16/(3N^2)$. The admissible ceiling forbids the unbounded curvature that appears at $r = 0$ in classical general relativity or at the Big Bang, provided the configuration remains in the admissible cone. Black hole and cosmological singularities are replaced by finite, multi-layered torsional configurations in which repulsive layers ($J < 0$) halt collapse before a true singularity can form.

9.2 Absence of Event Horizons

In the continuum limit, light rays follow null geodesics that terminate at an event horizon. In the discrete dual spacetime, however, light rays encounter only a finite number of torsional layers. Resonance tunneling through these layers remains possible at all depths. The classical Kerr outer horizon is reinterpreted as a torsional transition shell rather than a one-way membrane. Observers falling inward experience a sequence of attractive and repulsive torsional regions but never encounter a true horizon. This resolves the black hole information paradox and the firewall problem without invoking exotic Planck-scale physics.

9.3 Strong-Field Deviations

Gravitational time dilation and redshift receive discrete corrections. The proper time interval measured by a stationary observer at radial coordinate r is

$$\frac{\Delta\tau}{\tau} = \sqrt{1 - \frac{2GM}{c^2 r}}\left[1 + \mathcal{O}\left(\frac{\ell_P^2}{r^2}\right)\right].$$

For the Earth's core the correction is of order 10^{-20}, for neutron star surfaces approximately 10^{-10}, and near Sgr A* of order 10^{-5}. These deviations are in principle detectable with next-generation atomic clocks and very-long-baseline interferometry.

9.4 Gravitational-Wave Echoes

After a binary merger, the remnant torsion star possesses a stratified interior with alternating attractive and repulsive layers. Gravitational waves partially reflect at each torsional interface, producing a train of echoes. The time delay between successive echoes is

$$\Delta t \sim \frac{GM}{c^3} \approx 10^{-5}\left(\frac{M}{10M_\odot}\right)\ \text{s}.$$

The echoes carry a characteristic phase modulation determined by the dual-rotor spectrum. Detection of such echoes by LISA or the Einstein Telescope would constitute direct evidence for the discrete torsional structure of strong gravity.

9.5 Cosmological Implications

The early universe is described as a single primordial torsion star. The Planck-era torsional mismatch drives a brief de Sitter-like expansion without requiring an inflaton field. Reheating occurs through layer transitions that convert torsional energy into radiation. The resulting primordial power spectrum is naturally scale-invariant, arising from the discrete rotor modes rather than from quantum fluctuations of a scalar field. Dark matter is unnecessary: galactic rotation curves are explained by residual baryonic torsional misalignment at galactic scales. Dark energy emerges as the minimum non-zero vacuum torsional rigidity $J_{\text{vac}} > 0$ induced by quantum rotor fluctuations, yielding a cosmological constant of order $\Lambda \sim \ell_P^{-2}$.

All of the above predictions are falsifiable within the next decade. Non-observation of strong-field deviations at the predicted level, absence of gravitational-wave echoes, or detection of a true event horizon would rule out the framework. Conversely, positive detection of any of these signatures would confirm that spacetime is discrete, particle-local, and doubly temporal at the most fundamental level.

10 Comparison with Other Quantum Gravity Approaches

The double spacetime theory occupies a distinctive position among quantum gravity candidates. Most frameworks quantize or discretize an a priori continuum geometry. The results collected here are algebraic identities in $\mathrm{Cl}(3,1)$ and $G(3,1,1)$, not a derivation of quantum mechanics and gravity from that algebra. Below we compare the programme with existing approaches and record why it does not identify with a ten-dimensional string.

10.1 Loop Quantum Gravity

Loop quantum gravity (LQG) discretizes spacetime via spin networks and quantizes area and volume operators while retaining a background manifold and Riemannian curvature as fundamental. Singularities are resolved by a minimal eigenvalue of the area operator, yet the continuum limit remains challenging and the dynamics are encoded in complicated Hamiltonian constraints.

In contrast, the double spacetime theory imposes discreteness directly on the dual-rotor angles $\phi_a, \theta_a \in (2\pi/N)\mathbb{Z}$, without ever introducing a background manifold. Curvature is absent; gravity is torsional mismatch in the Lie algebra $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$. The reconstruction problem of LQG—recovering a smooth spacetime from spin networks—does not arise, because macroscopic spacetime emerges collectively from the torsional alignment of neighboring particle-intrinsic dual rotors.

10.2 String Theory

String theory unifies forces by embedding gravity in a ten- or eleven-dimensional spacetime populated by vibrating strings and requiring supersymmetry. While mathematically elegant, it suffers from a vast landscape of vacua and lacks unique low-energy predictions.

The algebras $\mathrm{Cl}(3,1)$ and $\mathrm{Cl}(9,1)$ are not isomorphic, and the dual spinor space $\mathbb{C}^2$ is not the sixteen-real-dimensional Majorana–Weyl module of ten-dimensional spacetime. Unification inside four-dimensional $\mathrm{Cl}(3,1)$ is a statement about one algebra, not a proof that extra dimensions are physically absent. The dual map is an internal automorphism; it is not spacetime supersymmetry. Gravity, electromagnetism, and the nuclear forces as modes of a single rotor algebra remain hypotheses.

No Calabi–Yau compactification is used; compactness of the dual sector at the particle level is a geometric hypothesis, not a string compactification.

10.3 Canonical and Ashtekar Approaches

The Ashtekar reformulation casts general relativity in terms of self-dual connections, enabling a Yang–Mills-like quantization. However, it inherits the continuum singularities of classical gravity and must confront the Wheeler–DeWitt equation with its notorious factor-ordering and regularization problems.

In the present framework the connection is eliminated entirely. Parallel transport is replaced by rotor exponentiation, and the Killing form on the torsional bivector replaces the curvature scalar. The resulting theory is constraint-free at the classical level and quantizes naturally via the finite rotor group, avoiding the Wheeler–DeWitt equation altogether.

10.4 Emergent Gravity and Causal Sets

Emergent gravity approaches treat spacetime as a thermodynamic or entropic phenomenon, while causal set theory discretizes Lorentzian order. Both frameworks struggle to recover the exact Einstein equations and face fermion doubling or chirality issues.

The double spacetime theory recovers the Einstein equations precisely through its equivalence with teleparallel gravity, while a commuting projector in $G(3,1,1)$ generates an eight-dimensional left ideal that splits by the null generator and is not read as an irreducible fermion. Discreteness is algebraic (Diophantine constraints on rotor angles) rather than set-theoretic, and causality is preserved by the lightlike $J = 0$ resonance condition. No thermodynamic analogy is invoked; the emergence of spacetime is a collective effect of torsional alignment among particle-intrinsic dual rotors.

In summary, existing quantum gravity programs extend or quantize the continuum manifold while retaining curvature as fundamental. The present paper records algebraic identities in $\mathrm{Cl}(3,1)$ and $G(3,1,1)$ and discards several identifications that do not hold. It does not derive spacetime, all interactions, or a controllable theory of gravity from those identities.

11 Experimental Testability from the Quantum Perspective

Although the double spacetime theory is fundamentally algebraic, its quantum sector makes concrete experimental predictions that can be tested with current or near-future technology. The mismatch scalar $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ depends only on the relative rotor. Dual-only motion cannot raise J; it conserves $J + M$, so every dual-only shield from a pure usual seed doubles unsigned mass. Shielding is a global budget, $\sum\alpha \leq 3\pi/4$, not an isotropic per-axis condition: a single boosted axis can still be dual-only shielded by parking dual rapidity on a silent axis. Repulsion requires a strictly negative wall. The vacuum $J = M = 0$ is mixed-reachable and dual-only unreachable except from the origin. The dual-only increment of the dual angles is helicity-even in J: a circular Faraday wave does not displace the mean, while a circular potential drops it ponderomotively and leaves the cone from a vanishing dual seed. Whether laboratory fields supply a helicity-odd dual-channel force remains a conjecture.

11.1 Theoretical Coupling Mechanism

A circularly polarised electromagnetic wave is null at every phase, with vanishing four-phase mean of every linear Faraday coefficient. Superposing that wave onto the mismatch does not shift mean J. Among the six classical Faraday quadratics, the unique helicity-odd combination on that wave is the Poynting component $(E \times B)_z = \sigma E_0^2$. The same dual-only increment, read as $\theta \mapsto \theta + eA$, drops mean J by the helicity-even ponderomotive $\frac{1}{2}e^2\langle A^2\rangle$ and is not Maxwell-gauge invariant. Helicity drive of the particle mismatch is neither that increment nor a local scalar of F. The gauge-invariant electromagnetic coupling that the algebra does supply is the minimal coupling of Section 5, in which the magnetic field acts on the dual spinor through the dual generators. The working model takes the Poynting bilinear as the candidate source of a dual-phase drive, with right-handed polarisation ($\sigma = +1$) reinforcing attractive mismatch ($J > 0$) and left-handed polarisation ($\sigma = -1$) opposing it. The effective interaction is written

$$H_{\text{int}} = \kappa\sigma E_0^2 \sum_a \theta_a \Gamma_a,$$

where E_0 is the electric-field amplitude and κ is a material-dependent coupling, neither of which is derived from the dual-rotor algebra. The phase-lock

$$\dot{\theta} = \kappa\sigma E_0^2 \sin(\omega t - \theta + \delta_\sigma)$$

is extra structure. The written particle action, given a dual-channel acceleration u, makes the mismatch a double integrator $\ddot{\delta} = -u$ and supplies no restoring frequency. Dual-only kinematics is a constrained slice of that action; without the constraint, both sectors move.

11.2 Proposed Near-Term Experiments

The most accessible platform exploits collective coherence in superconductors or Bose–Einstein condensates, where macroscopic numbers of particles ($N \sim 10^{26}$) can lock their dual rotors into a common phase. A concrete proposal is as follows:

- Suspend a microgram-scale superconducting test mass inside an asymmetric THz resonator (quantum cascade laser or free-electron laser cavity) with precise circular-polarization control.
- Monitor the apparent weight or inertial mass via a high-precision torsion balance or superconducting gravimeter while sweeping the frequency and helicity.
- Look for helicity-dependent changes in effective mass. A resonant peak at a proposed dual-rotor scale $\theta \sim c/(2\pi r_{\text{comp}})$ would test the extra phase-lock structure, not the dual-only geometry of J, and not the helicity-even ponderomotive of the dual-only increment.

Predicted effect sizes, if the coupling exists, are $\Delta m/m \sim 10^{-2}$–10^{-5} at achievable intensities of 10^{12}–10^{15} W/m^2. The signature is distinctive: the effect must reverse sign when the polarization helicity is flipped and must vanish for linearly polarized or unpolarized radiation.

11.3 Falsifiability and Caveats

A null result at the 10^{-6} level after a systematic frequency and helicity scan would constrain the dual-rotor coupling strength, not the admissible-cone kinematics of J. Positive detection—particularly a helicity-reversible mass shift—would constitute strong evidence that gravity and inertia are torsional phenomena originating in dual spacetime. A resonance peak would further constrain the working oscillator or phase-lock, which the written action does not force.

The theory makes no claim of immediate technological breakthrough. The proposals above are modest, table-top experiments whose primary purpose is to test the hypothesis that dual-spacetime degrees of freedom couple measurably to laboratory electromagnetic fields. Success at this level would open the door to larger-scale gravitational engineering; failure would simply bound the parameters of the coupling model.

11.4 Broader Implications

If dual-rotor control proves feasible, the same mechanism offers pathways to nuclear transmutation and radioactivity neutralization by driving torsional layer transitions inside atomic nuclei. These applications, however, lie beyond the scope of the present quantum-gravity test program and will be addressed in future work.

In summary, dual-only kinematics already delimit when J can be shielded or reversed, at what unsigned-mass cost, and that the dual budget may be routed across silent axes. Whether nature supplies a laboratory dual force—and whether gravity is therefore a controllable torsional degree of freedom—is the experimental question of the coming decade.

12 The algebraic picture

The quantum sector of double spacetime theory is the kinematics of a compact dual spacetime already present in the 16-real-dimensional algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$. Gravity is the dynamical cost of breaking the parallelism between the usual and dual rotors that every massive particle carries. No background manifold is quantized. The Hilbert-layer pairing is treated in [1].

The dual sector is elliptic. Its generators square to -1, satisfy $IJ = K$, and act on $\mathbb{C}^2$ by $-i\sigma_a$. At every dual rapidity the dual rotor is an SU(2) matrix, unitary of determinant one; relative dual rotors remain in the group. The inverse is the opposite rapidity, so a relative dual rotor is a product whose character records the alignment of the two dual axes, with a traceless noncommutative remainder. Compactness of the dual spacetime is compatible with this matrix representation. This is the spin-1/2 kinematics of the theory: a representation on $\mathbb{C}^2$, not a Riemannian identification of the rotor manifold with the three-sphere, and not a derivation of position operators. Usual and dual generators commute on a common axis and need not commute off-axis. With the usual factor written first, every total rotor factors uniquely as $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$ in this representation, and the dual factors are exactly the total rotors that fix the rest momentum. The little group of a massive particle, which governs its spin, is the dual sector.

The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators of the same algebra. They are not the hyperbolic bivectors used to build square-$+1$ projectors: γ^0 squares to -1, while the generator j of P_{spin} squares to $+1$. Fermionic kinematics therefore sits inside the algebra. A commuting square-$+1$ projector generates an eight-dimensional left ideal in $G(3,1,1)$ that splits as a four-dimensional Minkowski core plus a null copy; the copy is a proper submodule, so the ideal is reducible as a PGA-module. The real dimension matches $\mathbb{C}^4$ by that doubling, not by identification with an irreducible Dirac module. A dual spinor is already a $\mathbb{C}^2$ vector for the cyclic generators, and the Dirac carrier is assembled from it on the mass shell. The Clifford square of a four-momentum is the Minkowski quadratic $Q(p)$. The momentum-space Dirac equation is the first-order factor of that quadratic. On the mass shell a dual spinor u determines a solution; at rest momentum this is the pair $\Psi = (u, iu)$. Dual rotation of u stays on that shell. A longitudinal boost of the rest pair is the same Weyl construction after a chiral factor along the boost axis, which commutes with dual rotation about that axis and need not commute with a perpendicular dual rotation. The rest-frame overlap of two such solutions that differ by a dual rotation is twice the dual-rotor amplitude, and a longitudinal boost of rapidity α multiplies it by $\cosh\alpha$. The mass bilinear $\Psi^\dagger(-i\gamma^0)\Phi$ equals that rest-frame overlap and is unchanged by the boost. The factor $\cosh\alpha$ is the time component of the Dirac current $i\bar{\Psi}\Phi(\cosh\alpha, 0, 0, -\sinh\alpha)$, whose Clifford contraction with the boosted momentum is im times the bilinear. The eigenrays of σ_z label the spin along that momentum.

The boost preserves each ray, the mass bilinear and the current between opposite rays vanish, and a perpendicular dual rotation of angle π exchanges the rays. These boosts and rotations are instances of the covariance of the momentum-space equation under every total rotor, which preserves the mass bilinear and the contraction of the current with a co-transformed vector. The momentum of a solution is carried by the usual factor, and exactly one usual rotor reaches it from rest. Two usual rotors along perpendicular axes leave a nontrivial dual rotation about the third axis, the Thomas–Wigner rotation. The current of two solutions at one momentum is the raised momentum times the mass bilinear, so the Euclidean overlap is p_0/m times that bilinear. Under $p_\mu \mapsto -i\partial_\mu$ the momentum-space equation becomes the spacetime Dirac equation $\gamma^\mu \partial_\mu \Psi = -m\Psi$. On-shell plane waves and their superpositions solve it, and the current of any two solutions of one mass is conserved. Its time component gives the nonnegative density $\Psi^\dagger \Psi$, which dual rotors leave unchanged and the usual factor scales by p_0/m. Acting on spacetime points as well, a total rotor carries solutions to solutions and plane waves to plane waves of the transformed momentum. Minimal coupling to an electromagnetic potential is gauge covariant, places the mass shell at the kinetic momentum, and keeps the current gauge invariant and conserved. The commutator of covariant derivatives is the field strength, so every solution of the coupled equation obeys a second-order equation. Its magnetic term is a dual generator acting identically on both Weyl blocks, with gyromagnetic ratio 2; its electric term is a usual generator acting oppositely on them. At zero charge it is the Klein–Gordon equation. A spin Hamiltonian linear in the Pauli matrices evolves a dual spinor by a dual rotor, so magnetic spin precession is dual rotation. None of this derives the value of m from torsion. Equal electromagnetic phases order their interference nodes by rapidity, so every common phase that decreases strictly with distance places those nodes inward, and they are distinct from the outward Bohr radii $n^2 a_0$.

Gravity is the mismatch of the two rotors, measured by the parameter-space scalar $J = \frac{1}{2}\sum_a (\alpha_a^2 - \beta_a^2)$. On admissible configurations $|J| \leq 3\pi^2/8$, or equivalently $|J_{\mathrm{norm}}| \leq 1$. Compactness does not force an algebraic ceiling of order N^{-2}: the discrete quantity $\varepsilon_N = 16/(3N^2)$ is a lower bound on nonzero normalized height. Vacuum suppression, if it occurs, must come from a dynamical cancellation between the hyperbolic and cyclic sectors, whose Killing form takes opposite signs. The Killing self-pairing itself is indefinite of signature $(3,3)$ and is not a Born probability [1]; the Born density is the time component of the Dirac current. Identification of J with the teleparallel scalar, and variational equivalence with the Einstein–Hilbert action, are taken from the literature after a rotor-to-tetrad dictionary is fixed [7]. The dual map $X \mapsto Xi$ is an internal automorphism of $\mathrm{Cl}(3,1)$. It is not spacetime supersymmetry of a ten-dimensional string: $\mathrm{Cl}(3,1) \ncong \mathrm{Cl}(9,1)$, and the dual spinor space is not the Majorana–Weyl module of $\mathrm{Cl}(9,1)$.

The written opposite-sign particle action does not close the dynamical bridge from this kinematics to quantum theory. Its Euler–Lagrange system frees the mismatch and sources the common rapidity: $\ddot{\delta} = 0$, $\ddot{\sigma} = -2m\delta$. The integral is an affine lag and a cubic common rapidity, and the Jacobi integral along that motion reduces to initial data. The integral is indefinite, so conservation does not restore a frequency. An oscillator appears only if both kinetic signs are taken positive, and then at $\sqrt{2m}$, which meets the Compton frequency m only at the isolated values $m = 0$ and $m = 2$. Identifying the rate of free lag with the de Broglie phase remains a reading. The harmonic stiffness does not select a misalignment: a nonzero lag is inertial only where the slope of the potential vanishes. Measurement as synchronization, entanglement, and the path integral remain geometric hypotheses placed on this kinematics. They are not consequences of the algebra.

The continuum remains a classical approximation. The 16-dimensional algebra is the setting of the quantum sector, not a finished theory of quantum gravity.

13 Conclusions

The double spacetime theory supplies an algebraic setting for quantum gravity, not a completed derivation. Quantum kinematics is the compact dual sector of $\mathrm{Cl}(3,1)$; gravity is the mismatch of usual and dual rotors. Section 12 records what that algebra determines: the $\mathrm{SU}(2)$ dual rotor, the momentum-space Dirac equation solved on the mass shell by a dual spinor that recovers the rest pair (u, iu), the longitudinal boost of that pair, which leaves the mass bilinear invariant and multiplies the Euclidean overlap by $\cosh\alpha$ as the time component of the Dirac current, the spin along that momentum as an eigenray of σ_z, preserved by the boost and exchanged by a perpendicular dual rotation of angle π, the covariance of that equation under every total rotor, the unique factorization of a total rotor into a usual and a dual rotor, the dual sector as the little group of the rest momentum, the Thomas–Wigner rotation as the dual factor of perpendicular usual rotors, the Gordon identity that ties the current of two solutions

to their momentum and mass bilinear, the spacetime Dirac equation obtained by $p_\mu \mapsto -i\partial_\mu$, solved by on-shell plane waves and carrying a conserved current whose time component is the nonnegative Born density, the covariance of that equation under total rotors acting on spacetime points, its gauge-covariant minimal coupling with the mass shell at the kinetic momentum, the second-order equation in which the magnetic field acts through the dual generators with gyromagnetic ratio 2 and the electric field through the usual generators, spin precession in a magnetic field as a dual rotation, the admissible ceiling on J, and the inward order of equal-scale interference nodes. It does not determine a Compton oscillator from the written action, mass as torsion, or the outward Rydberg radii.

Predictions and laboratory proposals in the preceding sections remain conjectural, bounded by that ceiling. Dual-sector kinematics is the $\mathbb{C}^2$ representation described in [1]. The continuum remains a classical approximation. The 16-dimensional algebra is the setting, not a finished theory of quantum gravity.

References

[1] Anonymous, "Dual Spacetime 4-Valued Logic (D4L): Amplitudes, Four States, Geometric Interference, and Dual Hilbert Space" (2026). `https://github.com/hypernumbernet/DstDiophantine/blob/main/papers/dst-4-valued-logic.tex`.

[2] Anonymous, "Gravity as Torsion between Dual Spacetime: A Biquaternionic Reformulation of General Relativity" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/double-spacetime-theory.pdf`.

[3] Anonymous, "De Broglie Waves as Dual-Rotor Phase Lag: Quantum Mechanics as Geometry of Particle-Intrinsic Dual Spacetime" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/dst-de-broglie-wave.pdf`.

[4] Anonymous, "Dual-Rotor Dynamics in Double Spacetime Theory: Written Action, Free Mismatch, and a Neighbouring Oscillator" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/dst-dynamic-equation.pdf`.

[5] Anonymous, "Discrete Dual Spacetime Model and the Diophantine Structure of Mass Spectra" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/discrete-dual-spacetime.pdf`.

[6] A. Einstein, "Riemann-Geometrie mit Aufrechterhaltung des Begriffes des Fernparallelismus," *Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl.*, 217–221 (1928).

[7] J. W. Maluf, "The teleparallel equivalent of general relativity," *Ann. Phys. (Berlin)* **525**, 339–359 (2013). doi:10.1002/andp.201200272

[8] R. Aldrovandi and J. G. Pereira, *Teleparallel Gravity: An Introduction*, Fundamental Theories of Physics **173**, Springer, Dordrecht (2013).

[9] S. Bahamonde, K. F. Dialektopoulos, C. Escamilla-Rivera, G. Farrugia, V. Gakis, M. Hendry, M. Hohmann, J. Levi Said, J. Mifsud and E. Di Valentino, "Teleparallel gravity: from theory to cosmology," *Rep. Prog. Phys.* **86**, 026901 (2023). arXiv:2106.13757 [gr-qc].

[10] P. R. Girard, "Einstein's equations and Clifford algebra," *Advances in Applied Clifford Algebras*, **9**, no. 2, 225–230 (1999). doi:10.1007/BF03042377

[11] D. Hestenes, "Spacetime physics with geometric algebra," *Am. J. Phys.* **71**, 691–714 (2003).

[12] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, Cambridge (2003).

[13] P. Lounesto, *Clifford Algebras and Spinors*, 2nd ed., London Mathematical Society Lecture Note Series **286**, Cambridge University Press, Cambridge (2001).

[14] D. Hestenes, "Real Spinor Fields," *J. Math. Phys.* **8**, 798–808 (1967).

[15] P. A. M. Dirac, "The quantum theory of the electron," *Proc. R. Soc. Lond. A* **117**, 610–624 (1928).

[16] T. D. Newton and E. P. Wigner, "Localized states for elementary systems," *Rev. Mod. Phys.* **21**, 400–406 (1949).

[17] P. W. Higgs, "Broken symmetries and the masses of gauge bosons," *Phys. Rev. Lett.* **13**, 508–509 (1964).

[18] F. Englert and R. Brout, "Broken symmetry and the mass of gauge vector mesons," *Phys. Rev. Lett.* **13**, 321–323 (1964).

[19] C. Rovelli, *Quantum Gravity*, Cambridge University Press, Cambridge (2004).

[20] A. Ashtekar and J. Lewandowski, "Background independent quantum gravity: a status report," *Class. Quantum Grav.* **21**, R53–R152 (2004).

[21] A. Ashtekar, "New variables for classical and quantum gravity," *Phys. Rev. Lett.* **57**, 2244–2247 (1986).

[22] B. S. DeWitt, "Quantum theory of gravity. I. The canonical theory," *Phys. Rev.* **160**, 1113–1148 (1967).

[23] J. Polchinski, *String Theory*, Vols. I and II, Cambridge University Press, Cambridge (1998).

[24] M. B. Green, J. H. Schwarz and E. Witten, *Superstring Theory*, Vols. 1 and 2, Cambridge University Press, Cambridge (1987).

[25] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521–524 (1987).

[26] J. Ambjørn, J. Jurkiewicz and R. Loll, "Reconstructing the universe," *Phys. Rev. D* **72**, 064014 (2005).

[27] M. Niedermaier and M. Reuter, "The asymptotic safety scenario in quantum gravity," *Living Rev. Relativ.* **9**, 5 (2006).

[28] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260–1263 (1995).

[29] E. P. Verlinde, "On the origin of gravity and the laws of Newton," *J. High Energy Phys.* **04**, 029 (2011).

[30] S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199–220 (1975).

[31] A. Almheiri, D. Marolf, J. Polchinski and J. Sully, "Black holes: complementarity or firewalls?" *J. High Energy Phys.* **02**, 062 (2013).

[32] V. Cardoso and P. Pani, "Testing the nature of dark compact objects: a status report," *Living Rev. Relativ.* **22**, 4 (2019).

[33] J. Abedi, H. Dykaar and N. Afshordi, "Echoes from the Abyss: Tentative evidence for Planck-scale structure at black hole horizons," *Phys. Rev. D* **96**, 082004 (2017).

[34] The Event Horizon Telescope Collaboration, "First M87 Event Horizon Telescope results. I. The shadow of the supermassive black hole," *Astrophys. J. Lett.* **875**, L1 (2019).

[35] The Event Horizon Telescope Collaboration, "First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way," *Astrophys. J. Lett.* **930**, L12 (2022).

[36] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.* **116**, 061102 (2016).

[37] P. Amaro-Seoane *et al.*, "Laser Interferometer Space Antenna," arXiv:1702.00786 [astro-ph.IM] (2017).

[38] M. Punturo *et al.*, "The Einstein Telescope: a third-generation gravitational wave observatory," *Class. Quantum Grav.* **27**, 194002 (2010).

[39] A. H. Guth, "Inflationary universe: a possible solution to the horizon and flatness problems," *Phys. Rev. D* **23**, 347–356 (1981).

[40] A. A. Starobinsky, "A new type of isotropic cosmological models without singularity," *Phys. Lett. B* **91**, 99–102 (1980).

[41] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1–23 (1989).

[42] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics Physique Fizika* **1**, 195–200 (1964).

[43] A. Einstein, B. Podolsky and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?" *Phys. Rev.* **47**, 777–780 (1935).